$$\Delta U = -2.1 \times 10^{-12}\ J$$

(b) $$\Delta KE = -\Delta U$$

$$\Delta KE = +2.1 \times 10^{-12}\ J$$

P.4 **What must be the magnitude of an isolated +ve point charge from the ele potential at 15 cm from the charge to be 120 V?**

Sol.

DATA

$V = 120V$

$r = 15 \times 10^{-2}\ m$

$q = ?$

CALCULATIONS

$$V = \frac{kq}{r}$$

$$q = \frac{Vr}{k} = \frac{120 \times 15 \times 10^{-2}}{9 \times 10^{9}}$$

$$q = 2 \times 10^{-9}\ C$$

P.5 **What is the electric potential at the surface of a gold nucleus? The radi 7×10^{-15} m and atomic no. = 79 ?**

Sol.

DATA

$q = 79e = 79 \times 1.6 \times 10^{-19}\ C$

$r = 7 \times 10^{-15}\ m$

$V = ?$

CALCULATIONS

$$V = \frac{kq}{r}$$

$$V = \frac{9 \times 10^{9} \times 79 \times 1.6 \times 10^{-19}}{7 \times 10^{-15}}$$

$$V = 1.6 \times 10^{7} \text{ Volt}$$

$$\frac{}{dr}$$

$$E = \frac{+k\,qZ}{(R^2 + Z^2)^{3/2}}$$

P.16 Calculate the radial potential gradient in V/m at the surface of the gold nucleus of radius 7.0×10^{-15} m and 79 protons.

Sol.

DATA

$$r = 7.0 \times 10^{-15}\,m$$

$$q = 79e = 79 \times 1.6 \times 10^{-19}\,C$$

$$\frac{dV}{dr} = ?$$

CALCULATIONS

$$V = \frac{k\,q}{r}$$

$$\frac{dV}{dr} = \frac{-k\,q}{r^2} = \frac{-9 \times 10^9 \times 79 \times 1.6 \times 10^{-19}}{(7.0 \times 10^{-15})^2}$$

$$\frac{dV}{dr} = -2.3 \times 10^{21}\,V/m$$

P.17 A charge of 15 nC can be produced by simple rubbing. To what potential would such a charge raise an isolated conducing sphere of 16 cm radius.

Sol.

DATA

$$q = 15 \times 10^{-9}\,C$$

$$r = 16 \times 10^{-2}\,m$$

$$V = ?$$

CALCULATIONS

$$V = \frac{k\,q}{r}$$

$$V = \frac{9 \times 10^{9} \times 15 \times 10^{-9}}{16 \times 10^{-2}} = 843.7 \text{ Volt}$$

(a) the charge (b) the charge density on the surface

$$\sigma = \frac{3.6 \times 10^{-9}}{4\pi \times (15.2 \times 10^{-2})^2} = 1.25 \times 10^{-8}\ C/m^2$$

P.19 Potential at the center of a uniformly charged circular disk of radius 3.5 c 550 V. What is the total charge in the disc.

Sol.

DATA

$r = 3.5\ cm = 3.5 \times 10^{-2}\ m$

$V = 550\ V$

$q = ?$

CALCULATIONS

$$V = \frac{kq}{r}$$

$$q = \frac{Vr}{k} = \frac{550 \times 3.5 \times 10^{-2}}{9 \times 10^{9}}$$

$$q = 2 \times 10^{-9}\ C$$

$$V = \left(\frac{2 \times 9 \times 10^{\ } \times 1.76 \times 10^{\ } \times 1.6 \times 10^{\ }}{9.1 \times 10^{-31} \times 1.22 \times 10^{-2}}\right)$$

$$V = 21367 \text{ m/s}$$

P.11 A space shuttle moves through the dilute ionized gas of the earth ionosphere its potential is typically changes by –1.0 V before it complete one revolution. By assuming that the shuttle is a sphere of radius 10 cm, estimate the amount of charge it collects.

Sol.

DATA

$r = 10 \times 10^{-2}$ m

$V = -1.0$ Volt

$q = ?$

CALCULATIONS

$$V = \frac{kq}{r}$$

$$q = \frac{Vr}{k}$$

$$q = \frac{-1 \times 10 \times 10^{-2}}{9 \times 10^{9}} = -1.11 \times 10^{-11} \text{ C}$$